

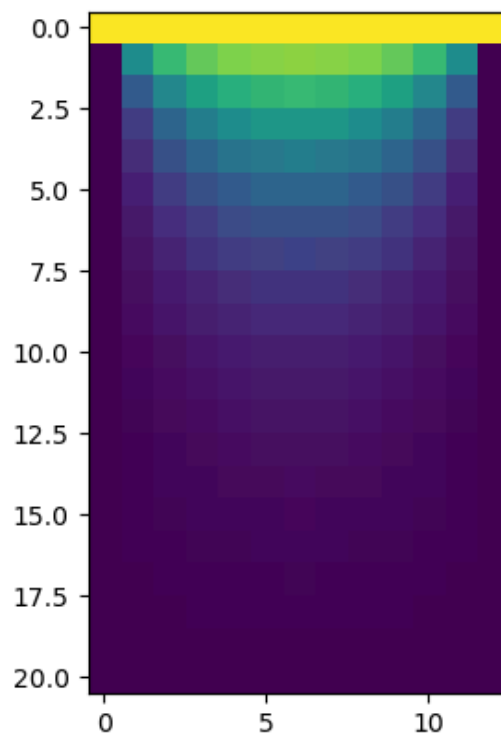
day 33 | 251114 F

```
In [37]: df = pd.read_excel('../data/laplace.xlsx', header=None)
         f = np.asarray(df)

         fig0, ax0 = plt.subplots()
         ax0.imshow(f)
```

Out[37]: <matplotlib.image.AxesImage at 0x75fc26de63f0>

Figure



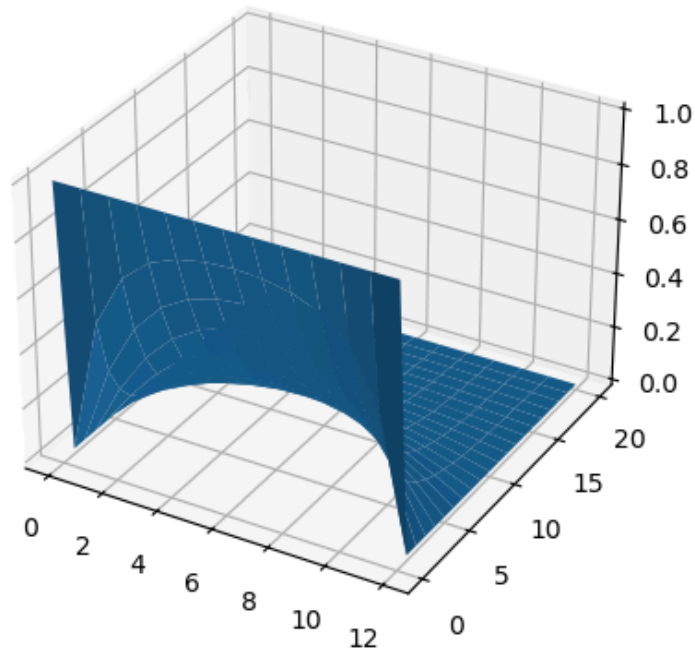
```
In [38]: length = f.shape[1]
         width = f.shape[0]
```

```
In [39]: x, y = np.meshgrid(np.arange(length), np.arange(width))
```

```
In [40]: fig1, ax1 = plt.subplots(subplot_kw={"projection": "3d"})
         ax1.plot_surface(x, y, f)
```

Out[40]: <mpl_toolkits.mplot3d.art3d.Poly3DCollection at 0x75fc26cef7d0>

Figure



```
In [41]: M = 100 # grid size
         V = 1.0 # voltage

         target = 2**-.16
```

```
In [42]: phi = np.zeros([M+1, M+1], float)
         phi[0, :] = V

         phiprime = np.empty([M+1,M+1], float)
```

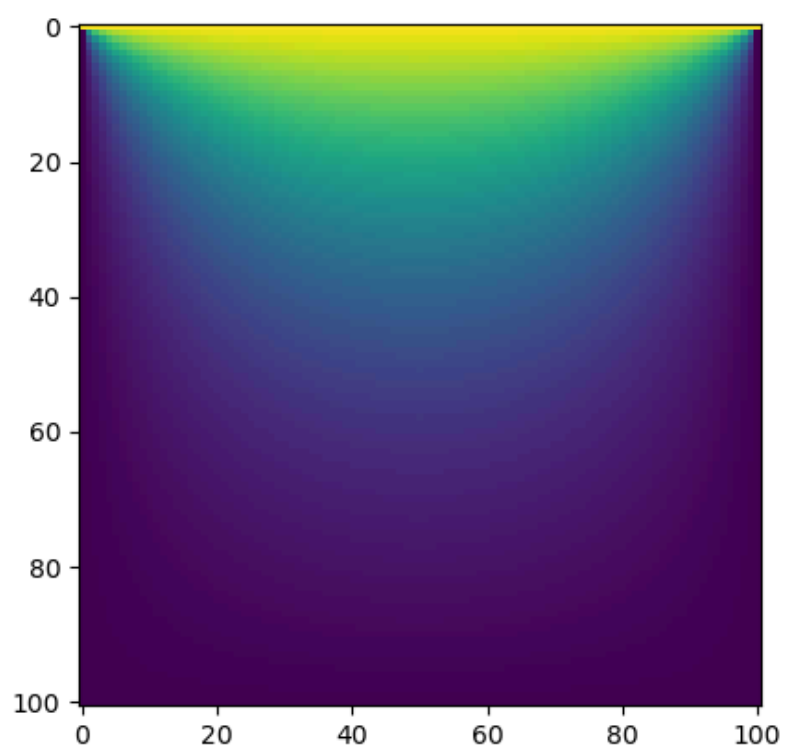
```
In [43]: delta = 1.0

         while delta>target:
             for i in range(M+1):
                 for j in range(M+1):
                     if i == 0 or i == M or j==0 or j==M:
                         phiprime[i,j] = phi[i,j]
                     else:
                         phiprime[i,j] = (phi[i+1,j]+phi[i-1,j]+phi[i,j+1]+phi[i,j-1])
                     delta = np.max(np.abs(phi-hiprime))
                 phi, phiprime = phiprime, phi
```

```
In [44]: fig2, ax2 = plt.subplots()
         ax2.imshow(phi)
```

```
Out[44]: <matplotlib.image.AxesImage at 0x75fc26b49580>
```

Figure



In []: